Tracking Control Three-Wheeled Omni-Directional Mobile Robot using PI Controller

*Son Duong Khanh and Thanh Tu Diep Cong
*Can Tho University of Technology, Vietnam
Faculty of Mechanical Engineering, Hochiminh City University of Technology, Vietnam
Email: tdcthanh@hcmut.edu.vn

Abstract: This paper presents a method of tracking control three-wheeled Omni-directional mobile robot (OMR) using PI controller based on a nonlinear dynamic robot model. With simple structure and easy to design, PI controller has highly effectiveness. The simulation results show that the controller has good performance, acceptable settling time and output closely follows the reference.
Keywords: Omni-Directional Mobile Robot, PI controller, MIMO.

1. INTRODUCTION

A three-wheeled Omni-directional mobile robot is a holonomic robot [1][2]. Using three Omni wheels, which are controlled by three independent motors, therefore, it can move and rotate in any direction even in the narrow space. The inherent agility of the three-wheeled OMR makes it widely studied for dynamic environment applications.

Because the model of three-wheeled OMR is a nonlinear MIMO system, the problem of accurately tracking complex trajectory is a difficult and received much research attention. This paper presents the application of PI controller for three-wheeled OMR. The results will be tested through simulations in Matlab.

2. METHODS

2.1. Literature review

The design of three-wheeled OMR controller has been studied based on different robot dynamics models. The controller in a Robocat system [3] was only based on kinematics. In additional, the trial-and-error process of tuning the PID controller gains is tedious [4]. Therefore, the controller response is not good, the error is still noticeable and the settling time is slow. In [4], the dynamics are simply a linear system due to ignoring the nonlinear coupling between the rotational and translational velocities. In this system, two independent PID controllers are designed to control the position and direction separately. In consequence, the controller does not appear to be effective when the robot has both the rotational and translational motion.

In [5], a detailed robot model has been developed as a nonlinear MIMO system, the paper developed a TLC (trajectory linearization control) control algorithm to control the robot to follow the trajectory. However, TLC method is quite difficult to find the inverse of the system. The performance of the controller also depends on the experience to select the damping ratio (ζ) and the natural frequency (ω).

This paper consists of three main parts. On the first part, the detail model of three-wheeled OMR is derived, including nonlinear elements and nonlinear coupling between them. The latter part introduce the design of PI controllers for tracking control three-wheeled OMR to follow the trajectory. It consists of an outer-loop (for kinematics) controller and an inner-loop (for dynamics) controller. Finally, the simulation verifies the result in part three.

2.2. Three-wheeled Omni-directional mobile robot

Robot model are derived based on some typical simplified assumptions. One assumption is that the wheels do not slip in the direction of traction force. Friction on the motor shaft and gearbox is viscous friction coefficient. All other friction of wheel that is not in the direction of the applied force are ignored. Electrical time constant of the motor is also ignored. The structure of the robot is presented in figure 1. It has three Omni-directional wheels, arranged 120 deg apart. Each wheel is driven by a separated DC motor.

We use two coordinate axes: body coordinate system {B} and {W} world coordinate system as shown in figure 1. The body coordinate system is fixed to the vehicle body on a moving robot with the origin located at the geometric center of the robot. The world frame coordinate system is fixed in space that the robot operates.

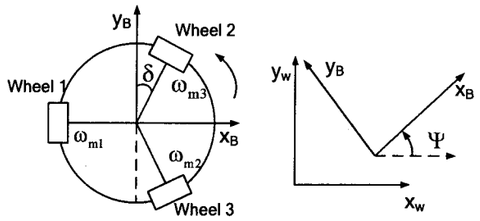


Fig.1 Body frame {B} and world frame {W}

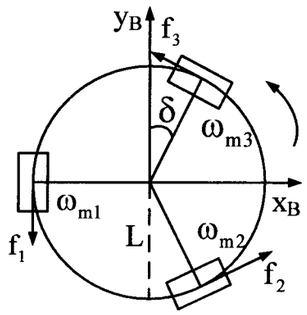


Fig.2 Force analysis.

With

v_ψ : Angular rate of body rotation;

v_x, v_y : Velocity component in the body frame;

f_1, f_2, f_3 : Traction force of each wheel;

E_1, E2, E3: Applied voltage on each motor;

$\omega_{m1}, \omega_{m2}, \omega_{m3}$: Motor shaft speed;

x, y : Robot location;

Ψ : Robot orientation angle;

m: Robot mass;

I_z: Robot moment inertia;

δ : 30^0;

L: Radius of robot body;

From figure 1, the robot kinematics equation is given by a coordinate transformation from the body frame {B} to the world frame {W}

$$\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\Psi}\end{bmatrix}=\begin{bmatrix}\cos\Psi(t) & -\sin\Psi(t) & 0\\ \sin\Psi(t) & \cos\Psi(t) & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}v_x\\ v_y\\ v_\psi\end{bmatrix} \quad (1)$$

The dynamics model is derived on the analysis of the forces from the body frame {B} in Figure 2 as follows

$$\begin{bmatrix}\dot{v}_x\\ \dot{v}_y\\ \dot{v}_\psi\end{bmatrix}=\begin{bmatrix}v_\psi v_y\\ -v_\psi v_x\\ 0\end{bmatrix}+H.B\begin{bmatrix}f_1\\ f_2\\ f_3\end{bmatrix} \quad (2)$$

With

$$H=\begin{bmatrix}\frac{1}{m} & 0 & 0\\ 0 & \frac{1}{m} & 0\\ 0 & 0 & \frac{1}{I_Z}\end{bmatrix} \quad (3)$$

$$B=\begin{bmatrix}0 & \cos(\delta) & -\cos(\delta)\\ -1 & -\sin(\delta) & \sin(\delta)\\ L & L & L\end{bmatrix}$$

From the robot geometry analysis we obtain

$$\begin{bmatrix}v_x\\ v_y\\ v_\psi\end{bmatrix}=(B^T)^{-1}\frac{R}{n}\begin{bmatrix}\omega_{m1}\\ \omega_{m2}\\ \omega_{m3}\end{bmatrix} \quad (4)$$

The dynamics of each DC motor is given by the following two equations

$$L_a\frac{di_a}{dt}+R_a i_a+k_3\omega_m=E \quad (5)$$

$$J_0\dot{\omega}_m+b_0\omega_m+\frac{Rf}{n}=k_2 i_a \quad (6)$$

Where

E: Armature voltage;

i_a: Armature current;

ω_m : Motor shaft speed;

L_a: Armature inductance;

R_a: Armature resistance;

k_3: Back emf constant;

k_2: Motor torque constant;

J_0: Combined inertia of the motor, gear train and wheel applied to the motor shaft;

b_0: Combined viscous-friction coefficient of the motor, gear and wheel;

f : Wheel traction force;

R: Wheel radius;

n: Gear ratio;

Electrical time constant of the motor is ignored, so equation (5) is rewritten as follows

$$i_a=\frac{1}{R_a}(u-k_3\omega_m) \quad (7)$$

And

$$J_0\begin{bmatrix}\dot{\omega}_{m1}\\ \dot{\omega}_{m2}\\ \dot{\omega}_{m3}\end{bmatrix}+b_0\begin{bmatrix}\omega_{m1}\\ \omega_{m2}\\ \omega_{m3}\end{bmatrix}+\frac{R}{n}\begin{bmatrix}f_1\\ f_2\\ f_3\end{bmatrix}=\frac{k_2}{R_a}\begin{bmatrix}f_1\\ f_2\\ f_3\end{bmatrix}-\frac{k_2k_3}{R_a}\begin{bmatrix}\omega_{m1}\\ \omega_{m2}\\ \omega_{m3}\end{bmatrix} \quad (8)$$

Combining (2), (4), (8) the dynamics model of the robot in body frame {B} with the motor voltage E1, E2, E3 is the control input signal as follows

$$\begin{bmatrix}\dot{v}_x\\ \dot{v}_y\\ \dot{v}_\psi\end{bmatrix}=G^{-1}\begin{bmatrix}v_\psi v_x\\ -v_\psi v_y\\ 0\end{bmatrix}-G^{-1}HBB^T\left(\frac{k_2k_3}{R_a}+b_0\right)\frac{n^2}{R^2}\begin{bmatrix}v_x\\ v_y\\ v_\psi\end{bmatrix}$$

$$+G^{-1}HB\frac{k_2 n}{R.R_a}\begin{bmatrix}E_1\\ E_2\\ E_3\end{bmatrix} \quad (9)$$

with $G=(I+HBB^T\frac{n^2J_0}{R^2})$

Equations (1) and (9) show that the robot moves as a nonlinear MIMO system with coupling between the rotational and translational velocities.

2.3. Proportional– Integral controller

Proportional– Integral (PI) controller is used widely in many applications in industries because of the simple structure as well as good performance [6].

The basic structure of the PI controller is described in the flowing equations and as show in figure 3.

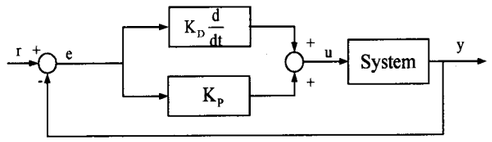


Fig.3 PI controller

$$u(t)=K_p e(t)+K_i\int e(\tau)d\tau$$
$$e(t)=r(t)-y(t) \quad (10)$$

Where r(t) is the reference input, u(t) is the output, e is the tracking error, K_p is the proportional gain, K_I is the integral gain of controller.

Characteristics of PI controllers: The proportional controller reduces rising time but not eliminates error in steady state. The integral control eliminates error in steady state, but can reduce the quality of the response.

To tune in the parameters required for stability and good transient performance can be a challenge . There are several prescriptive rules used in PID tuning. A method is that proposed by Ziegler and Nichols in the 1940's and it is used in this parper.

The PI controller tuning method Zeigler - Nichols

Turn the controller to P-only mode, i.e. turn the Integral modes off.

Turn the controller gain, K_c, up slowly and observe the output response. Note that this requires changing K_c in step increments and waiting for a steady state in the output, before another change in K_c is implemented.

When a value of K_c results in a sustained periodic oscillation in the output (or close to it), mark this critical value

of K_c as K_u, the ultimate gain. Also, measure the period of oscillation, P_u, referred to as the ultimate period.

Using the values of the ultimate gain, K_u, and the ultimate period, P_u, Ziegler and Nichols prescribes the following values for K_c, τ_I is desired: $K_c = K_u / 2.2$, $\tau_I = P_u / 1.2$.

3. SIMULATION RESULTS

In this section, results of simulation are presented. The PI controller for three-wheeled OMR built with Matlab Simulink and its simulation diagram is show in figure 4 with the coefficient of the PI outer loop is $K_P = 10, K_I = 6$ and PI inter loop is $K_P = 400, K_I = 320$.

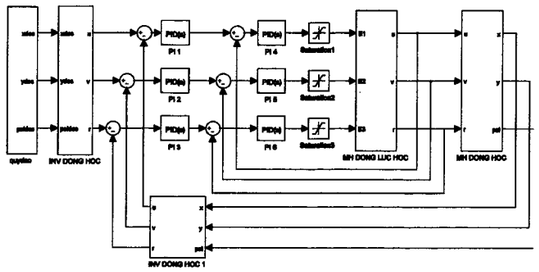


Fig.4 Simulation diagram

3.1. Simulation with circle trajectory

In this test, the robot is commanded to follow a circle trajectory with diameter is 1.4m. The robot orientation is fixed. Circle trajectory is described in the following equations

$$x = 0.7 - 0.7\cos(0.1t)$$
$$y = 0.7\sin(0.1t) \qquad (11)$$
$$\psi = 0$$

The tracking response, tracking error of x, y and Ψ are represented in Figure 6, 7 and 8. Output of the PI controller closely follows the reference, tracking error of x, y is 0.008m, tracking error of ψ is very small (0.0003rad).

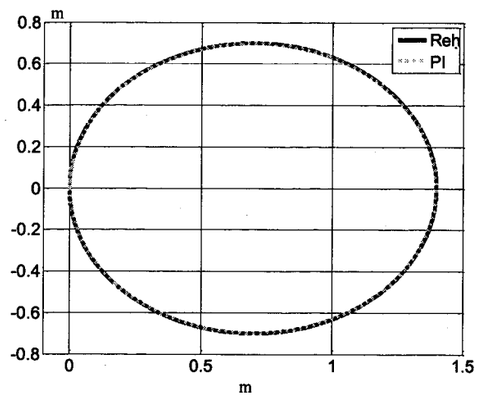
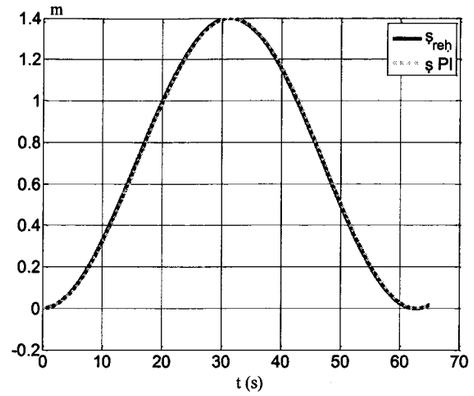


Fig.5 Robot trajectory



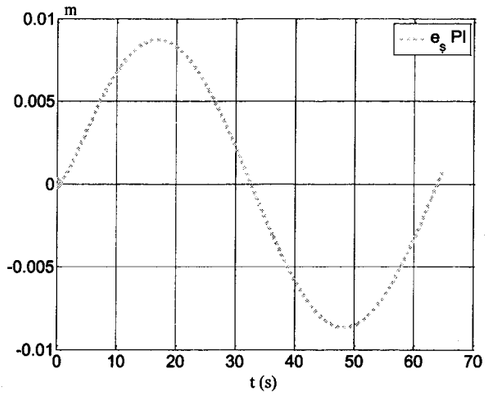
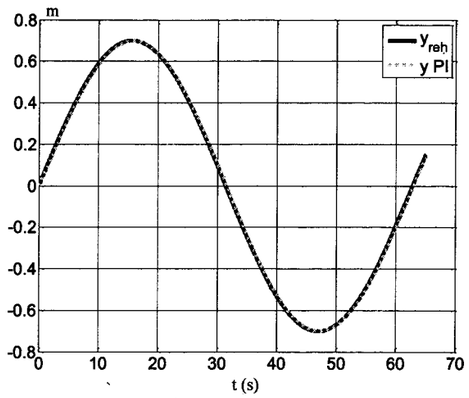


Fig.6 Tracking response and tracking error of x



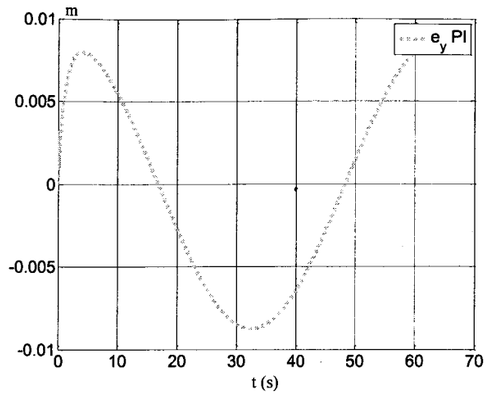


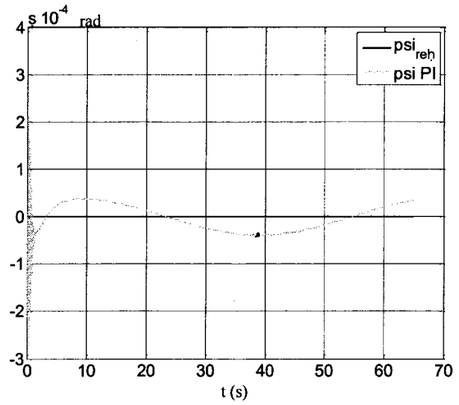
Fig.7 Tracking response and tracking error of y

3.2. Simulation with rose trajectory

In this test, the robot is commanded to follow a rose trajectory. The robot orientation is fixed. Rose trajectory is described in the following equations

$$x = \cos(2*0.05*t)*\sin(0.05*t)$$
$$y = \cos(2*0.05*t)*\cos(0.05*t)-1 \qquad (12)$$
$$\psi = 0$$

The tracking response, tracking error of x, y and Ψ are represented in Figure 10, 11 and 12. The trajectory of robot is closely follows the reference, tracking error is small. Therefore, the PI controller has good performance.



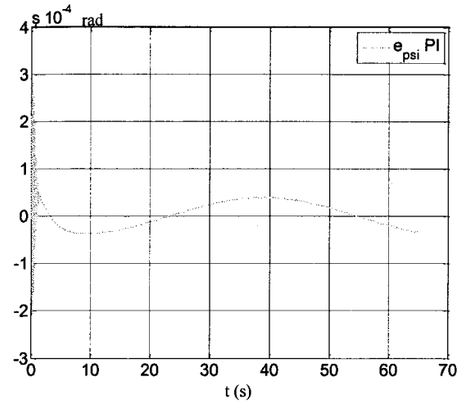


Figure 8: Tracking response and tracking error of Ψ

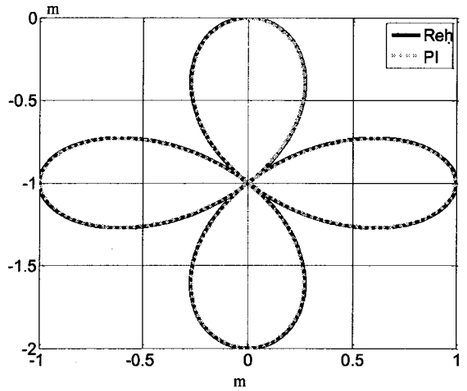
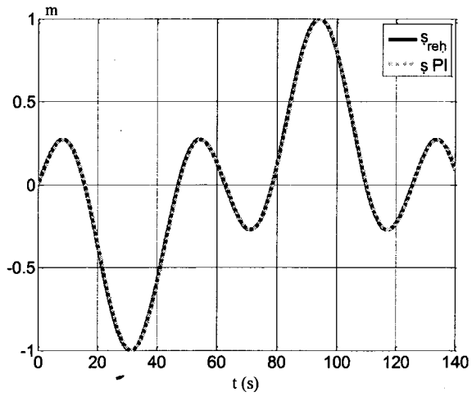


Fig.9 Robot rose trajectory



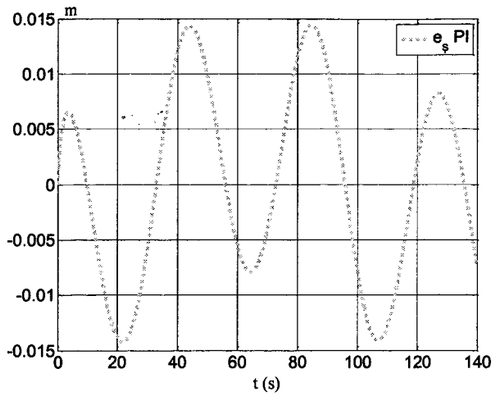
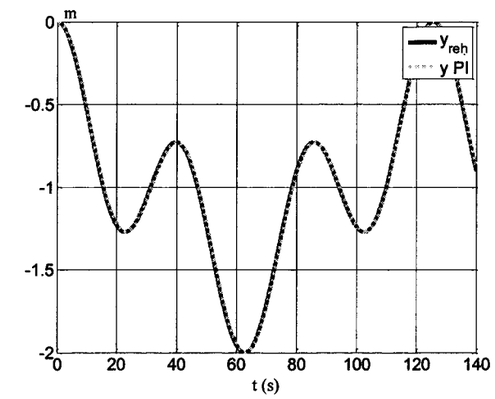


Fig.10 Tracking response and tracking error of x



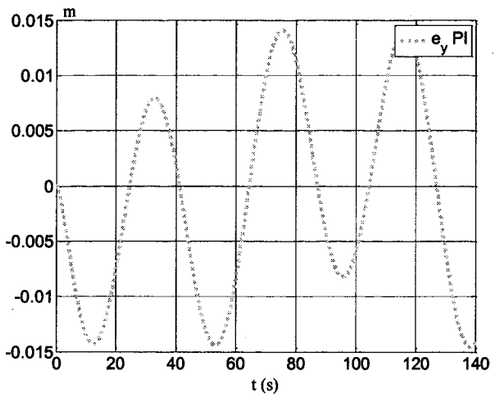
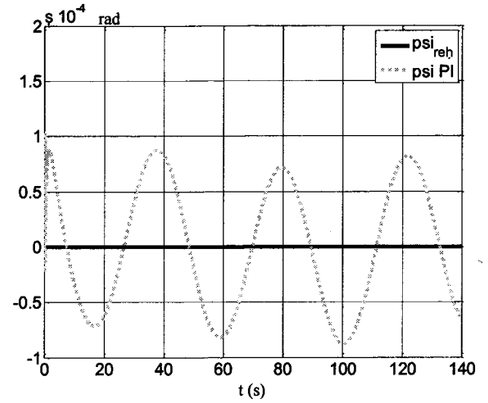


Fig.11 Tracking response and tracking error of y



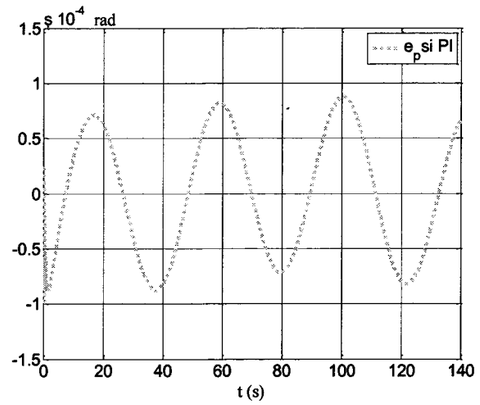


Fig.12 Tracking response and tracking error of Ψ 4.

4. CONCLUSION

This paper introduces the design of a controller for three-wheeled Omni-directional mobile robot using PI controller. The many advantages of this method include simplicity to design controllers, providing relatively good performance. Simulation results and experiment tests show that designed controllers are good performance for controlling the robot follow a complex command trajectory. The effectiveness of the controller depends on the fine-tuning of parameters K_p and K_I. However, there is no specific method to obtain the optimal value. They can be achieved only through the simulation and evaluation process. This problem will be addressed by the self-tuning fuzzy PI controller.

REFERENCES

[1] H.A. Samani, A. Abdollahi, H. Ostadi, S.Z. Rad, Design and development of a comprehensive omni directional soccer robot, International Journal of Advanced Robotic Systems 1 (3) (2004) 191–200.

[2] I.E. Paromtchik, U. Rembold, A practical approach to motion generation and control for an omnidirectional mobile robot, in: Proceedings of 1994 IEEE International Conference on Robotics and Automation, 1994, pp. 191–200.

[3] M. West, H. Asada, Design of a holonomic omnidirectional vehicle, in: Proceedings of the 1992 IEEE International Conference on Robotics and Automation, 1992, pp. 97–103.

[4] F.G. Pin, S.M. Killough, A new family of Omni-directional mobile robot-directional and holonomic wheeled platforms for mobile robots, IEEE ransactions on Robotics and Automation 10 (2) (1994) 480–489.

[5] Y. Liu, X. Wu, J. Zhu, J. Lew, "Omni directional mobile robot control by trajectory linearization", in , Denver, CO., 2003, pp. 3423-Proc. American Control Conference 3428.

[6] J. Wu, Dynamic path planning of an omni-directional robot in a dynamic environment, Ph.D. Dissertation, Ohio University, Athens, OH, 2005.